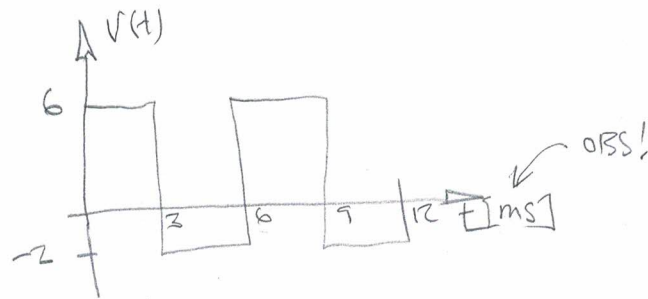


4.30)



sök kvoten $V_{\text{medel}}/V_{\text{rms}}$ för vågformen ovan

Lösning:

$$V_{\text{medel}} = \frac{1}{6 \cdot 10^{-3}} \left[\int_0^{3 \cdot 10^{-3}} 6 dt + \int_{3 \cdot 10^{-3}}^{6 \cdot 10^{-3}} (-2) dt \right] =$$

$$= \frac{1}{6 \cdot 10^{-3}} (3 \cdot 10^{-3} \cdot 6 - 2 \cdot (6 \cdot 10^{-3} - 3 \cdot 10^{-3})) =$$

$$= 2V$$

$$V_{\text{rms}}^2 = \frac{1}{6 \cdot 10^{-3}} \left[\int_0^{3 \cdot 10^{-3}} 6^2 dt + \int_{3 \cdot 10^{-3}}^{6 \cdot 10^{-3}} (-2)^2 dt \right] =$$

$$= \frac{1}{6 \cdot 10^{-3}} [6^2 (3 \cdot 10^{-3}) + 2^2 (3 \cdot 10^{-3})] =$$

$$= \frac{40}{6 \cdot 10^{-3}} \cdot 3 \cdot 10^{-3} = 20V \Rightarrow V_{\text{rms}} \approx 4.47 \text{ Volt}$$

kvoten blir

$$\frac{V_{\text{medel}}}{V_{\text{rms}}} \approx \frac{2}{4.47} \approx 0.447$$

SVAR:

$$\frac{V_{\text{medel}}}{V_{\text{rms}}} \approx 0.45$$